

Thematic Roles in Logical Form

A Detailed Explanatory Chapter

Introduction

Thematic roles (also called theta-roles) describe the semantic relationship between a verb and its arguments. While predicate logic can represent relational structure using ordered argument positions, it does not explicitly indicate the semantic function of each participant. For example, in the sentence “Rina broke the vase,” classical predicate logic represents this as $Break(r, v)$, but this does not tell us who performed the action and who was affected. Thematic roles provide this semantic information explicitly.

The development of event semantics, especially in the neo-Davidsonian tradition, allows thematic roles to be represented as independent predicates relating participants to event variables. This approach increases explanatory power because it separates the event from its participants and allows modifiers, alternations, and argument structure changes to be represented systematically.

This chapter develops a detailed understanding of thematic roles in logical form, explaining how they are encoded, how they interact with event structure, and how they help explain syntactic alternations.

1. From Predicate Positions to Thematic Roles

In classical predicate logic, a transitive verb like “break” is represented as:

$$Break(r, v)$$

Here, r denotes Rina and v denotes the vase. The first argument position corresponds to the subject, and the second to the object. However, nothing in this notation explicitly tells us that Rina is the agent and the vase is the affected entity.

In neo-Davidsonian semantics, we instead represent:

$$\exists e(Break(e) \wedge Agent(e, r) \wedge Theme(e, v))$$

Let us unpack this representation carefully.

- e is an event variable.
- $Break(e)$ says that e is a breaking event.
- $Agent(e, r)$ says that Rina performed the event.
- $Theme(e, v)$ says that the vase was the entity affected.

This decomposition separates the event predicate from the participant relations. It provides greater flexibility in representing argument structure.

2. Major Thematic Roles

Thematic roles describe the semantic function of participants in events.

2.1 Agent

The Agent is typically the intentional initiator of an event.

Example:

Rina opened the door.

Logical form:

$$\exists e(Open(e) \wedge Agent(e, r) \wedge Theme(e, d))$$

The predicate $Agent(e, r)$ indicates that Rina intentionally caused the event.

2.2 Theme

The Theme is the entity affected or undergoing change.

In the previous example, $Theme(e, d)$ indicates that the door is what underwent change.

Themes need not be objects of physical change. Consider:

Rina read the book.

$$\exists e(Read(e) \wedge Agent(e, r) \wedge Theme(e, b))$$

Here, the book is the Theme because it is the entity involved in the action.

2.3 Experienter

Experiencers undergo psychological states.

Rina fears spiders.

$$\exists s(Fear(s) \wedge Experienter(s, r) \wedge Theme(s, spiders))$$

Here, Rina is not performing an action but experiencing a state.

2.4 Goal and Source

Goal indicates endpoint; Source indicates starting point.

Rina went to Delhi.

$$\exists e(Go(e) \wedge Agent(e, r) \wedge Goal(e, delhi))$$

Rina came from Mumbai.

$$\exists e(Come(e) \wedge Agent(e, r) \wedge Source(e, mumbai))$$

These roles encode directional structure.

2.5 Instrument

Instrument denotes the means by which an event occurs.

Rina cut the bread with a knife.

$$\exists e(Cut(e) \wedge Agent(e, r) \wedge Theme(e, b) \wedge Instrument(e, k))$$

The knife is not an argument selected by the verb but an optional participant modifying the event.

2.6 Beneficiary

Beneficiary denotes the entity for whose benefit the event occurs.

Rina baked Amit a cake.

$$\exists e(Bake(e) \wedge Agent(e, r) \wedge Theme(e, c) \wedge Beneficiary(e, a))$$

3. Thematic Hierarchy

Thematic roles are often organized into a hierarchy:

$$Agent > Experiencer > Goal > Theme$$

This hierarchy predicts syntactic behavior. For example, Agents are most likely to appear as subjects. Themes typically appear as objects. This explains why in active sentences, the Agent surfaces as subject:

Rina broke the vase.

But in passive sentences:

The vase was broken by Rina.

Logical form remains:

$$\exists e(Break(e) \wedge Agent(e, r) \wedge Theme(e, v))$$

Thematic roles remain constant even though syntactic positions change.

4. Alternations and Logical Form

4.1 Active vs Passive

Active:

Rina broke the vase.

Passive:

The vase was broken by Rina.

Both share the same thematic structure:

$$\exists e(Break(e) \wedge Agent(e, r) \wedge Theme(e, v))$$

The difference lies in syntactic realization, not semantic structure.

4.2 Dative Alternation

Rina gave Amit a book.

Rina gave a book to Amit.

Logical form:

$$\exists e(Give(e) \wedge Agent(e, r) \wedge Theme(e, b) \wedge Goal(e, a))$$

Thematic roles remain constant across alternations.

5. Linking Theory

Linking theory explains how thematic roles map onto syntactic positions.

General tendencies:

- Agent $\rightarrow$ Subject
- Theme $\rightarrow$ Object
- Goal $\rightarrow$ Indirect Object / PP

These mappings are not arbitrary but reflect semantic prominence.

6. Compositional Derivation Example

Sentence:

Rina gave Amit a book.

Lexical entry:

$$give = \lambda z \lambda y \lambda x \lambda e (Give(e) \wedge Agent(e, x) \wedge Theme(e, z) \wedge Goal(e, y))$$

Step 1: Apply to object:

$$(\lambda z \dots)(b)$$

Step 2: Apply to indirect object:

$$(\lambda y \dots)(a)$$

Step 3: Apply to subject:

$$(\lambda x \dots)(r)$$

Step 4: Existential closure:

$$\exists e (Give(e) \wedge Agent(e, r) \wedge Theme(e, b) \wedge Goal(e, a))$$

Each step reflects function application.

Conclusion

Thematic roles provide the semantic architecture of argument structure. By introducing event variables and independent role predicates, neo-Davidsonian semantics separates event identity from participant relations. This approach explains alternations, linking patterns, and modifier attachment. Thematic roles are not merely descriptive labels but formal components of logical representation that reveal the underlying structure of verbal meaning.